

• h5i • PITCH

Is the Web App Your AI Built Really Secure?

AI agents now write and ship web apps. h5i is the workspace where developers and their AI check that what shipped is correct and secure.

■ SECURITY HAS TWO SIDES

Find bugs. Prove correctness.

Two sides of one coin.

01 · Red-teaming

Find bugs

Attack the running app the way a hacker would, and catch what nobody anticipated.

Expert hackers are expensive. Red-teaming AI can go rogue.

ONE



COIN

02 · Formal verification

Prove correctness

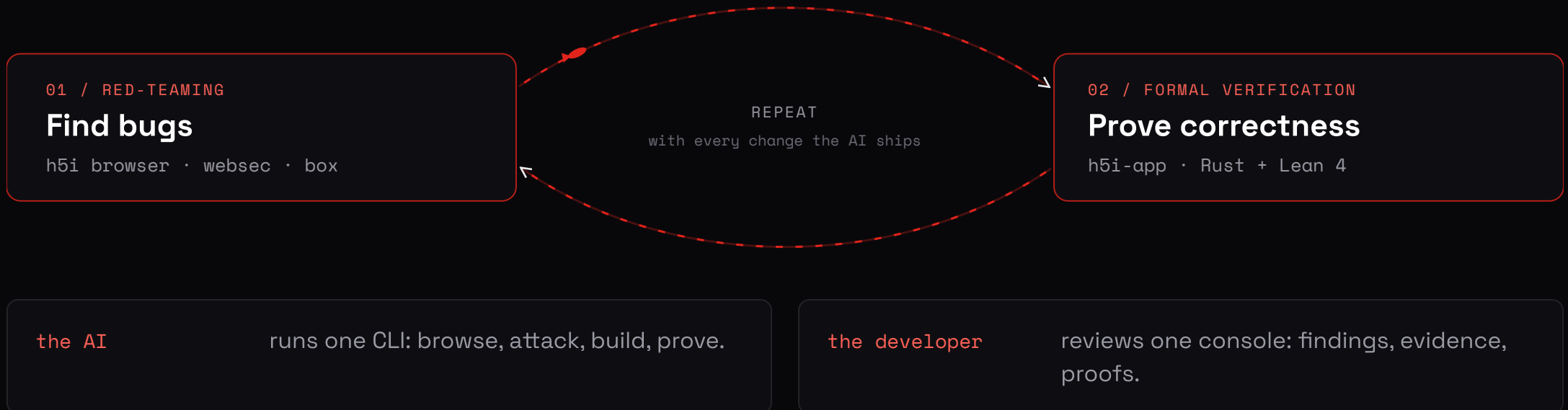
State the rules the logic must keep, and prove the code satisfies them.

Writing code is easy. Proving it correct is hard.

You need both, with every change the AI ships.

■ WHAT H5I IS

One workspace for both, for developers and their AI.



Your agent red-teams the running app.

It drives the app like a user and replays real requests like a hacker, inside limits you set. No framework migration required.

```
● ● ● the agent's session · an authorized target

$ h5i browser open https://shop.example --capture
$ h5i browser click @e7                                # "Orders", as Alice
$ h5i websec replay req_42 --set path=/api/order/1042
  res_43  200  {"id":1042,"owner":"Bob", ...}
$ h5i websec diff res_42 res_43
  owner: Alice → Bob                                # IDOR, evidence attached
```

Only the targets you allow

anything else is refused before the wire

Sandboxed workflow

supervised, container, or microVM

Every action recorded

browsing, traffic, refusals

State the rules.

Lean checks the code keeps them.

Write the logic on h5i-app, an Axum-based framework. Aeneas extracts it into Lean 4, where your theorems are checked.

RUST · APPLICATION KERNEL

```
board/kernel/src/lib.rs

fn delete(user: u64, s: &Snapshot,
          id: u64) -> Outcome {
  match find_post(&s.posts, id) {
    None => Err(Error::NotFound),
    Some(p) => {
      if p.author == user
      || is_moderator(&s.moderators, user)
      {
        Ok((one(Write::DelPost(id)),
            Reply::Done))
      } else {
        Err(Error::Forbidden)
      }
    }
  }
}
```

The logic your app runs

LEAN 4 · VERIFICATION

AENEAS
▶▶▶
EXTRACT

```
BoardKernel.lean EXTRACTED

def delete (user : U64) (s : Snapshot) (id : U64) := do
  let o ← find_post s.posts id
  match o with
  | none => ok (.Err .NotFound)
  | some p => if p.author = user then ...

Theorems.lean YOU SUPPLY

/-- Every successful write is allowed. -/
theorem authorized (a : Principal) (s : Snapshot)
  (c : Command) ws r (hinv : Inv (Snapshot.toSt s))
  (h : transition a s c = .ok (.Ok (ws, r))) :
  ∀ w ∈ ws.val,
    allowed (Snapshot.toSt s) a.user.val w := by ...
```

Checked against your specification

PROVEN PROPERTIES

✓ **authorized**
every write passes the policy

✓ **inv_preserved**
invariants survive every step

✓ **transition_total**
no panic, no overflow

Checked by Lean

A proof covers the stated property under stated assumptions. Red-teaming covers the rest.

■ WHY H5I

Other tools give you one side.
h5i gives you the coin.

Browser + proxy tools

Built for people. The agent juggles browser, proxy, and CA setup. No proofs.

Sandboxes

Limit what the agent can reach. No testing, no evidence, no proofs.

Proof assistants

Prove a model of the logic. Nothing connects it to the running app.

h5i

Red-team the running app and prove the logic, in one CLI the AI runs and one console you review.

■ TRACTION & USERS

Built in the open.

Built for teams that ship with AI.

670+

GitHub stars

60+

forks

20+

contributors

2.5k+

release downloads

GitHub · checked October 2, 2026 · [Repository and releases](#)

Product teams

shipping with AI, no security specialist

Security engineers

testing between assessments, with evidence

Rust teams

with rules that must never break

■ NEXT

Find bugs. Prove correctness. For every team that ships with AI.

Product

findings beside the proofs they touch, in one console

Adoption

10 design partners, then the first paid pilots

Open source

local-first, Apache 2.0, built with the community

Try it with your own AI agent: `curl -fsSL https://h5i.dev/install.sh | sh -s -- --websec --test`

h5i.dev

github.com/h5i-dev/h5i